



PLAYWRIGHT CAPSTONE PROJECT REPORT

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Project: IMDb Web Automation Testing Framework

GitHub Repository: https://github.com/ToshRohan/Capstone-Project_Wipro_Rohan-Tosh

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Executive Summary

This project presents an end-to-end test automation framework developed using **Playwright with JavaScript**. The framework is designed to validate multiple services of the **IMDb web application** through automated UI test cases, ensuring reliability, stability, and consistent user experience across different browsers.

The automation suite supports **cross-browser testing (Chromium, Firefox, and WebKit)** and is integrated with **GitHub Actions CI/CD pipeline** for continuous execution on every code push and pull request. The results are reported using **Allure Reports**, which provide detailed test execution insights including pass/fail status, logs, and execution history. The final reports are deployed using **GitHub Pages** for easy access and visualization.

The framework is built using the **Page Object Model (POM)** design pattern, which separates test logic from page-specific actions. This improves code reusability, scalability, and maintainability. Each IMDb service is structured as an independent module with dedicated test cases.

The project covers **seven IMDb services**, including modules such as Top 10 listings, Popular Celebrities, Movies/TV navigation, search functionality, and other key user-facing features.

Technology Stack

Playwright, JavaScript, Node.js, Git, GitHub, GitHub Actions, Allure Reports, GitHub Pages, VS Code.

Framework Architecture

The framework is based on the **Page Object Model (POM)** design pattern.

- Page classes handle locators and UI actions
- Test files focus only on assertions and test flow
- Utilities manage reusable functions and configurations
- CI pipeline handles automated execution and reporting

Modules Covered

- IMDb Home & Navigation
- Search Functionality

- Top 10 / Top Rated Content
- Popular Celebrities
- Movie & TV Show Listings
- Individual Title Details Pages

Service-wise Test Cases

Service 1 – IMDb Top 250 Movies

TC No.	Test Case	Business Value
TC01	Top 250 page opens	High
TC02	Movie list is visible	High
TC03	Ranking numbers are visible	High
TC04	Movie titles are displayed	High
TC05	Movie ratings are visible	High
TC06	Release year is displayed	Medium
TC07	Storyline is visible	Medium
TC08	Open movie details page	High
TC09	User reviews are visible	Medium
TC10	Ratings are valid	High
TC11	Search movie from Top 250	High
TC12	Page scroll works	Low
TC13	Page reload works properly	Low
TC14	No duplicate movie titles	High
TC15	Correct URL after opening movie	Medium

1.1 Non-Functional Requirements (NFR)

- **Performance:** The Top 250 page should load efficiently even with a large dataset of movies.
- **Browser Compatibility:** The service must function correctly on the latest version of Google Chrome.
- **UI Stability:** Page layout should remain consistent during scrolling and navigation.
- **Data Consistency:** Movie rankings, ratings, and titles must remain consistent across page refresh and navigation.

1.2 Test Strategy

A hybrid testing strategy is followed combining functional automation with UI validation using Playwright.

1.2.1 Manual Testing Scope

- **Exploratory Testing:** Manual verification of movie details and navigation flow from list to detail pages.
- **UI Validation:** Ensuring movie posters, titles, and ratings are properly displayed.
- **Negative Testing:** Checking behavior when invalid search inputs are provided or when network delays occur.

Service 2 – IMDb Popular Celebrities

TC No.	Test Case	Business Value
TC01	IMDb popular celebs page launched	High
TC02	Anne Hathaway profile visible	Medium
TC03	Navigation bar opens	High
TC04	Sally Field biography contains correct information	Medium
TC05	Search bar works	High
TC06	Sydney Sweeney awards	Medium
TC07	Christopher Nolan known for	Medium
TC08	More to explore section	Low
TC09	Sally Field profile loads correctly	Medium
TC10	Grid view works	Medium
TC11	Emily Blunt profile picture	Low
TC12	Most popular celebs heading	High
TC13	Detailed view	Medium
TC14	Search category	Medium
TC15	Celeb news page loads	Medium

2.1 Non-Functional Requirements (NFR)

- **Performance:** Celebrity pages should load without delay under normal network conditions.
- **Browser Compatibility:** Must work correctly on modern Google Chrome browsers.
- **Usability:** Navigation between celebrity profiles should be smooth and intuitive.
- **Content Accuracy:** Biography and profile details must be displayed correctly without missing data.

2.2 Test Strategy

A hybrid approach of automated and manual validation is used.

2.2.1 Manual Testing Scope

- **Exploratory Testing:** Checking celebrity profile navigation and content exploration.
- **UI Testing:** Validating profile images, grid layout, and page structure.
- **Negative Testing:** Verifying system behavior when invalid search queries are entered.

Service 3 – IMDb What To Watch

TC No.	Test Case	Business Value
TC01	Page loads with main UI components	High
TC02	Navigation menu is functional	High
TC03	Search bar accepts input	High
TC04	Recommendation content visible	High
TC05	Page reload stability	Low
TC06	Mobile view works	Medium

TC07	No error messages	High
TC08	Page scroll works	Low
TC09	Links are present and clickable	High
TC10	Search functionality works	High
TC11	Fan Favorites section is visible	Medium
TC12	Category selector works	Medium
TC13	Heading is visible	Low

3.1 Non-Functional Requirements (NFR)

- **Performance:** Recommendation content should load quickly without delay.
- **Responsiveness:** UI must adapt properly to different screen sizes (desktop and mobile).
- **Reliability:** Page should remain stable after refresh and repeated navigation.
- **Usability:** Search and filtering should be intuitive and user-friendly.

3.2 Test Strategy

Automation-focused testing supported with manual UI checks.

3.2.1 Manual Testing Scope

- **Exploratory Testing:** Checking recommendation flow and category filtering.
- **Usability Testing:** Ensuring easy navigation across different sections.
- **Negative Testing:** Verifying behavior when search inputs return no results.

Service 4 – IMDb Events

TC No.	Test Case	Business Value
TC01	Page loads with main UI components	High
TC02	12 Months Film Festival event loads correctly	Medium
TC03	Event links are functional	High
TC04	Search bar accepts input	High
TC05	Event details page loads	High
TC06	Mobile view works	Medium
TC07	Academy Awards winner info is correct	High
TC08	Mumbai Film Festival event page loads	Medium
TC09	Event details for Best Horror Baddy are visible	Medium
TC10	Cannes Film Festival event page loads	Medium

4.1 Non-Functional Requirements (NFR)

- **Performance:** Event pages should load efficiently with multiple event listings.
- **Data Accuracy:** Event details such as festivals and awards must be correct and updated.

- **Browser Compatibility:** Must function correctly on Chrome browser.
- **UI Consistency:** Layout should remain stable across different event pages.

4.2 Test Strategy

Combination of automated validation and manual verification.

4.2.1 Manual Testing Scope

- **Exploratory Testing:** Verifying event navigation and details accuracy.
- **UI Testing:** Checking event cards, headings, and layout structure.
- **Negative Testing:** Handling invalid search inputs or missing event data.

Service 5 – IMDb Satyajit Ray Profile

TC No.	Test Case	Business Value
TC01	Heading is visible	High
TC02	Personal details are visible correctly	High
TC03	Known for section is working correctly	High
TC04	Awards and nominations contain correct information	High
TC05	Biography section is visible	High
TC06	Writer section is visible	Medium
TC07	Trivia section is visible	Medium
TC08	Quotes section contains correct information	Medium
TC09	Photos section loads correctly	Medium
TC10	Composer section is visible	Medium

5.1 Non-Functional Requirements (NFR)

- **Content Accuracy:** Biography, awards, and filmography must be correctly displayed.
- **Performance:** Profile page should load without delay.
- **UI Stability:** Layout should remain consistent across sections like photos and trivia.
- **Browser Compatibility:** Must work properly on Chrome.

5.2 Test Strategy

Manual and automated validation of profile-based content.

5.2.1 Manual Testing Scope

- **Exploratory Testing:** Verifying different sections like awards, quotes, and biography.
- **Content Validation:** Checking correctness of profile data.
- **Negative Testing:** Testing missing or broken content sections.

Service 6 – IMDb Showtimes & Tickets

TC No.	Test Case	Business Value
TC01	Movie page loads successfully	High
TC02	Showtimes section is available on page or redirect exists	High
TC03	Showtimes button is clickable	High
TC04	Navigation to showtimes page works	High
TC05	Showtimes page content loads	High
TC06	Location prompt appears or handled	High
TC07	Cinema listings exist	High
TC08	Showtimes are visible	High
TC09	Ticket links are visible	High
TC10	External ticket redirect works	High
TC11	Page has no broken UI	High
TC12	Back navigation works	Medium
TC13	Page title is correct	Medium
TC14	Multiple sections load on showtimes page	Medium
TC15	No critical elements missing	High

6.1 Non-Functional Requirements (NFR)

- **Availability:** Showtimes and ticket links must always be accessible.
- **Performance:** Page should load cinema listings without delay.
- **Browser Compatibility:** Must work properly in Chrome browser.
- **External Navigation:** Ticket redirection should work without errors.

6.2 Test Strategy

Hybrid testing approach combining automation and manual verification.

6.2.1 Manual Testing Scope

- **Exploratory Testing:** Checking movie showtime flow and ticket navigation.
- **UI Testing:** Validating cinema listings, showtime tables, and buttons.
- **Negative Testing:** Handling missing location or unavailable showtimes scenarios.

Test Designing According to Expected Outcomes

Following the expected outcomes, the test cases were designed based on the functional flow of the IMDb web application. Each module represents a distinct service area of the platform, and test cases were created to validate both UI behavior and functional correctness.

The major modules covered in this project include:

- IMDb Top 250 Movies Page
- IMDb Popular Celebrities Page
- IMDb What To Watch Page
- IMDb Events Page
- IMDb Satyajit Ray Profile Page
- IMDb Showtimes & Tickets Page
- IMDb Homepage Add-on Components

Each module focuses on validating real user interactions such as navigation, search, content visibility, and page transitions.

1. Test Design Methodology

The automated test cases were designed to validate IMDb application workflows using Playwright with JavaScript. The primary objective was to ensure that each IMDb service behaves correctly under expected user actions.

The design approach followed these principles:

Path Verification (End-to-End Flow)

Tests were structured around key user journeys such as:

- Navigating from homepage to movie pages
- Searching for movies and celebrities
- Opening event pages and profiles
- Accessing showtimes and ticket links

Each test ensures that navigation and UI transitions work as expected.

Assertion-Based Validation

Each test case is built around assertions to verify:

- Element visibility (titles, images, sections)
- Functional correctness (navigation, search, links)
- Data accuracy (ratings, profiles, event details)

Requirement Traceability

Each test case maps back to IMDb functional requirements such as:

- Content discovery
- User navigation flows
- Media and profile viewing
- External link redirection

This ensures full traceability between requirements and automated test coverage.

2. Test Cases

IMDb Top 250 Movies Tests (15)

- **Top 250 page opens** – Validates that the Top 250 Movies page loads successfully.
- **Movie list is visible** – Ensures all ranked movies are displayed correctly.
- **Ranking numbers are visible** – Confirms correct ranking order display.
- **Movie titles are displayed** – Verifies visibility of movie names.
- **Movie ratings are visible** – Ensures IMDb ratings are displayed properly.
- **Release year is displayed** – Validates correct year information.
- **Storyline is visible** – Confirms movie details page shows storyline.
- **Open movie details page** – Checks navigation to individual movie pages.
- **User reviews are visible** – Ensures review section is accessible.
- **Ratings are valid** – Confirms rating format correctness.
- **Search movie from Top 250** – Validates search functionality.
- **Page scroll works** – Ensures smooth scrolling behavior.
- **Page reload works properly** – Confirms page stability after refresh.
- **No duplicate movie titles** – Ensures data consistency in listing.
- **Correct URL after opening movie** – Validates correct navigation URL.

IMDb Popular Celebrities Tests (15)

- **IMDb popular celebs page launched** – Confirms page loads successfully.
- **Anne Hathaway profile visible** – Validates celebrity profile visibility.
- **Navigation bar opens** – Ensures menu functionality works.
- **Sally Field biography contains correct information** – Validates profile accuracy.
- **Search bar works** – Confirms search functionality.
- **Sydney Sweeney awards** – Verifies awards section display.
- **Christopher Nolan known for** – Validates “Known For” section.
- **More to explore section** – Ensures recommendation content is visible.
- **Sally Field profile loads correctly** – Validates profile navigation.
- **Grid view works** – Confirms layout rendering.
- **Emily Blunt profile picture** – Ensures image visibility.
- **Most popular celebs heading** – Validates page heading display.
- **Detailed view** – Confirms extended profile information access.
- **Search category** – Ensures category filtering works.
- **Celeb news page loads** – Validates news page navigation.

IMDb What To Watch Tests (13)

- **Page loads with main UI components** – Validates full page rendering.
- **Navigation menu is functional** – Ensures menu navigation works.

- **Search bar accepts input** – Confirms search field usability.
 - **Recommendation content visible** – Validates content rendering.
 - **Page reload stability** – Ensures page remains stable after refresh.
 - **Mobile view works** – Confirms responsive design behavior.
 - **No error messages** – Ensures clean execution without UI errors.
 - **Page scroll works** – Validates scrolling behavior.
 - **Links are present and clickable** – Ensures navigation links work.
 - **Search functionality works** – Confirms working search results.
 - **Fan Favorites section is visible** – Validates section rendering.
 - **Category selector works** – Ensures filtering functionality.
 - **Heading is visible** – Confirms page title visibility.
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IMDb Events Tests (10)

- **Page loads with main UI components** – Validates Events page rendering.
 - **12 Months Film Festival event loads correctly** – Confirms event data visibility.
 - **Event links are functional** – Ensures navigation works.
 - **Search bar accepts input** – Validates search functionality.
 - **Event details page loads** – Confirms correct redirection.
 - **Mobile view works** – Ensures responsive behavior.
 - **Academy Awards winner info is correct** – Validates award data accuracy.
 - **Mumbai Film Festival event page loads** – Confirms page navigation.
 - **Event details for Best Horror Baddy are visible** – Ensures content display.
 - **Cannes Film Festival event page loads** – Validates event accessibility.
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IMDb Satyajit Ray Profile Tests (10)

- **Heading is visible** – Confirms profile title display.
 - **Personal details are visible correctly** – Validates biography data.
 - **Known for section is working correctly** – Ensures filmography display.
 - **Awards and nominations contain correct information** – Validates awards data.
 - **Biography section is visible** – Confirms biography rendering.
 - **Writer section is visible** – Ensures writing credits display.
 - **Trivia section is visible** – Validates trivia content.
 - **Quotes section contains correct information** – Ensures quotes display.
 - **Photos section loads correctly** – Validates image gallery loading.
 - **Composer section is visible** – Confirms composer credits visibility.
-

IMDb Showtimes & Tickets Tests (15)

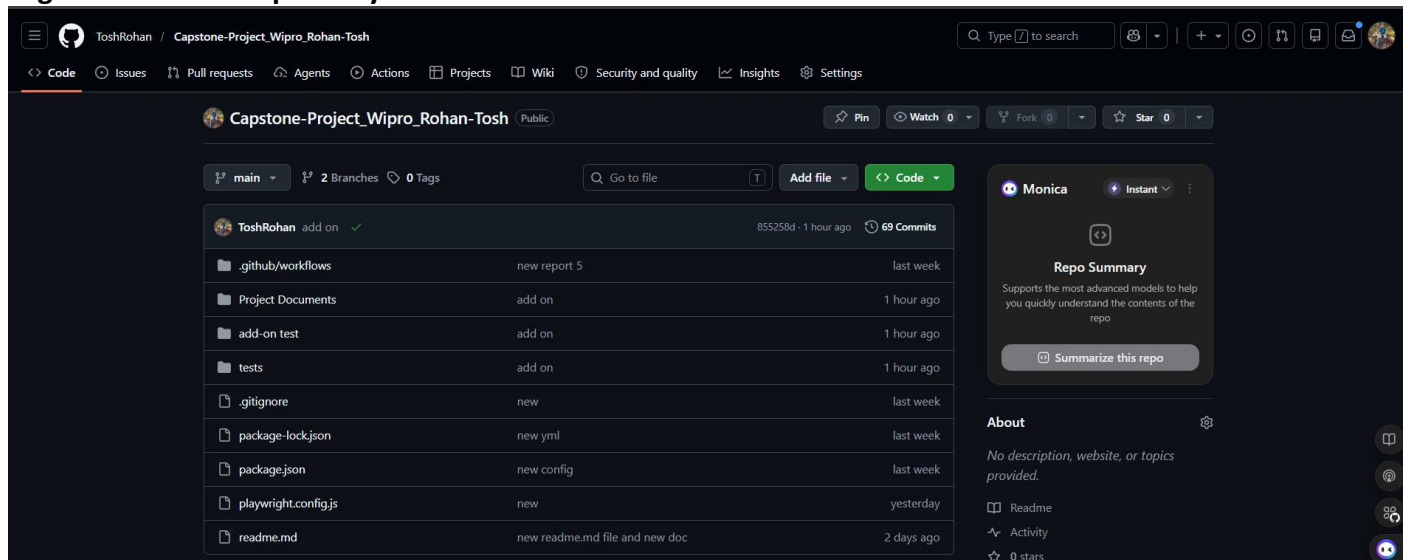
- **Movie page loads successfully** – Confirms movie page access.
- **Showtimes section is available on page or redirect exists** – Validates showtime availability.
- **Showtimes button is clickable** – Ensures button functionality.
- **Navigation to showtimes page works** – Confirms correct redirection.
- **Showtimes page content loads** – Validates schedule display.

- **Location prompt appears or handled** – Ensures location-based handling.
- **Cinema listings exist** – Confirms theater availability.
- **Showtimes are visible** – Validates schedule visibility.
- **Ticket links are visible** – Ensures booking links exist.
- **External ticket redirect works** – Confirms external navigation.
- **Page has no broken UI** – Ensures UI stability.
- **Back navigation works** – Validates browser navigation.
- **Page title is correct** – Confirms correct title rendering.
- **Multiple sections load on showtimes page** – Ensures full page rendering.
- **No critical elements missing** – Validates completeness of UI.

Execution Evidence and Reporting

GitHub Repository Structure

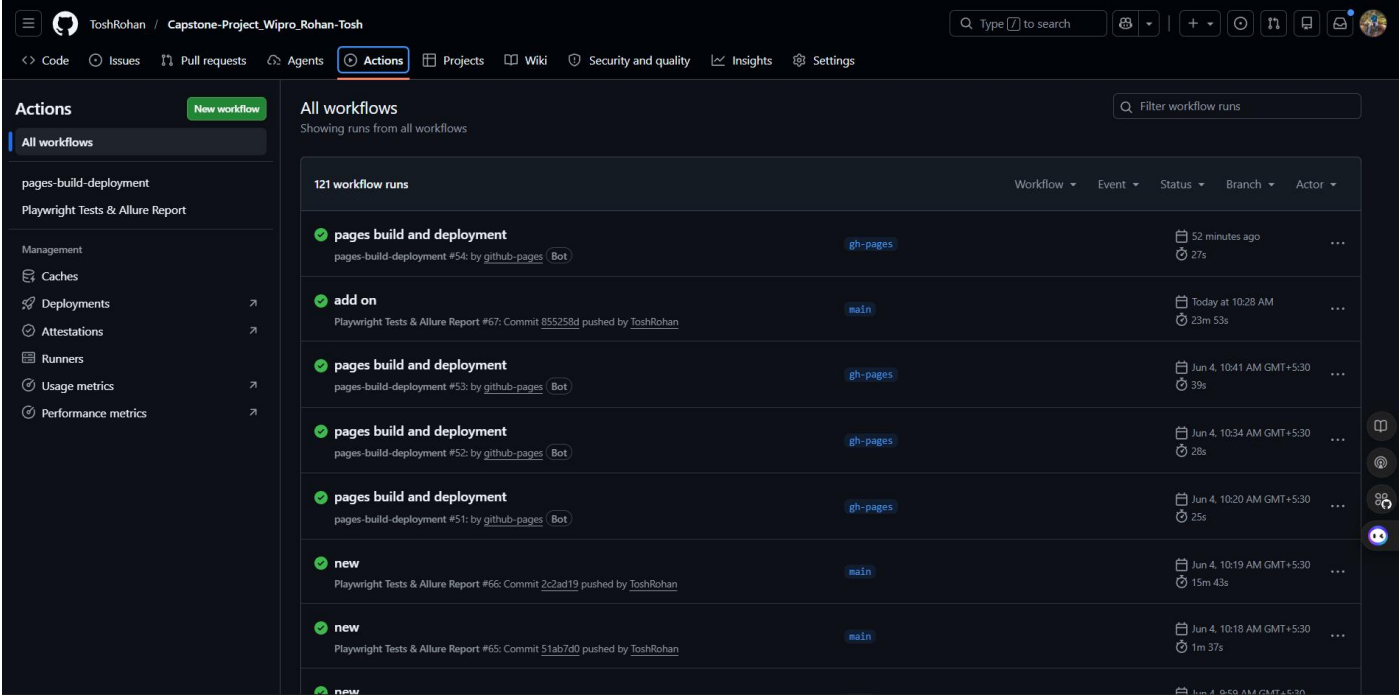
Figure 1: GitHub Repository Structure



This section presents the overall structure of the Playwright automation framework hosted on GitHub. It includes test directories, configuration files, CI/CD workflow files, and reporting folders.

GitHub Actions Workflow Execution

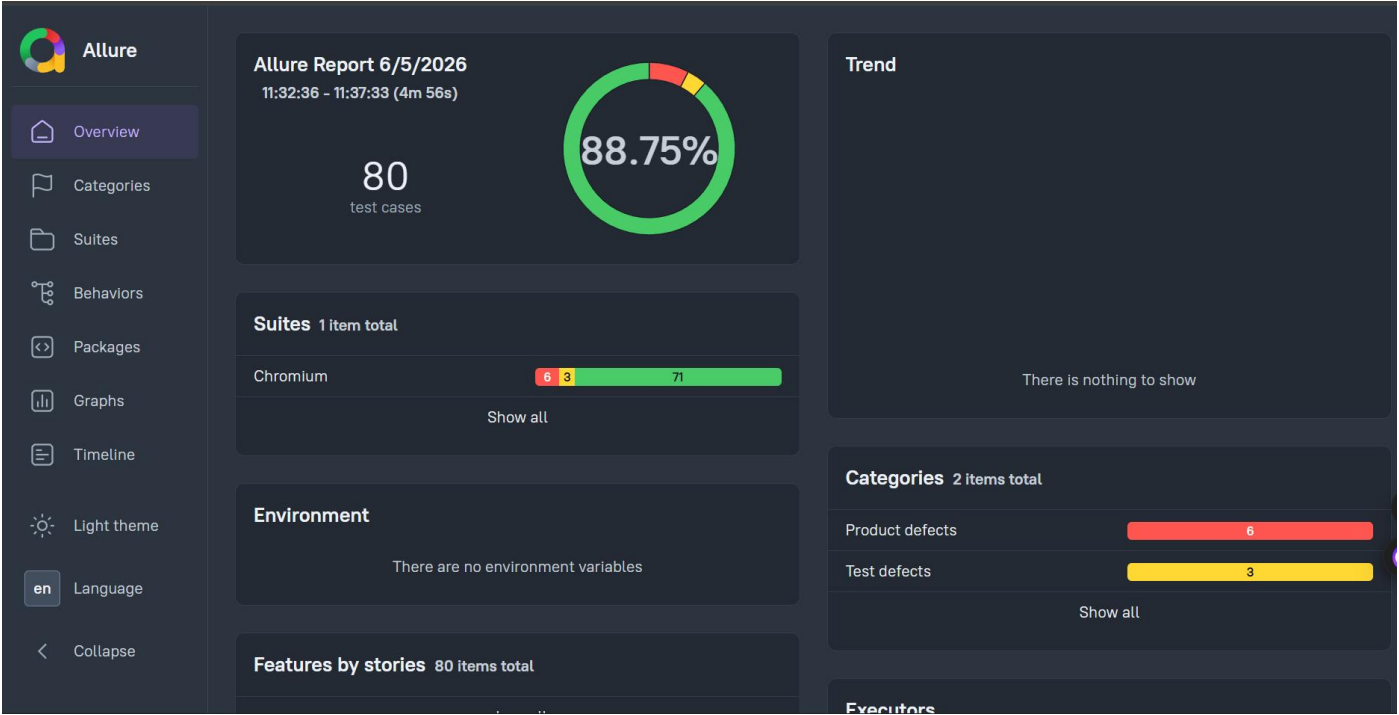
Figure 2: GitHub Actions Workflow Execution



This section demonstrates the successful execution of automated test cases using GitHub Actions CI/CD pipeline. Every push or pull request triggers the workflow which installs dependencies, executes tests, and generates reports.

Allure Dashboard Overview

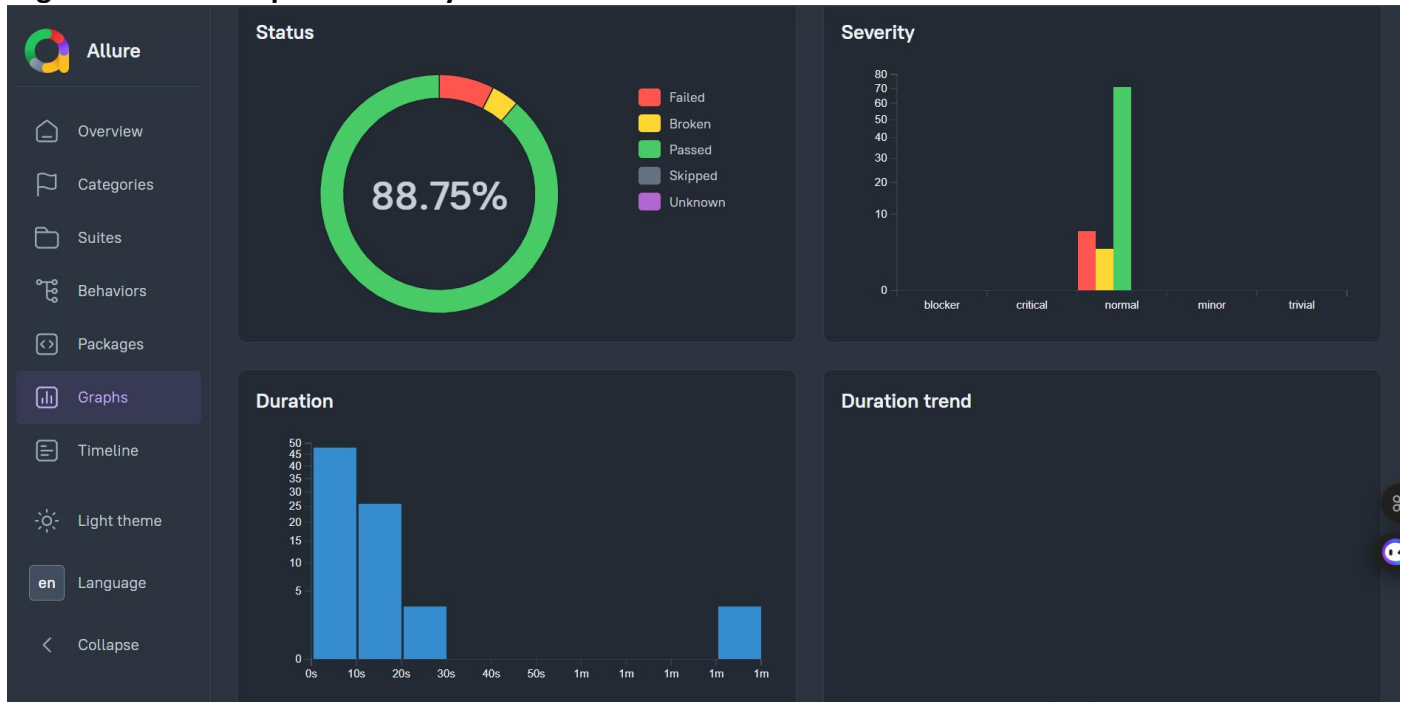
Figure 3: Allure Dashboard Overview



The Allure dashboard provides a detailed overview of test execution results including passed, failed, and skipped test cases. It also provides execution trends and historical data for analysis.

Allure Graphs and Analytics

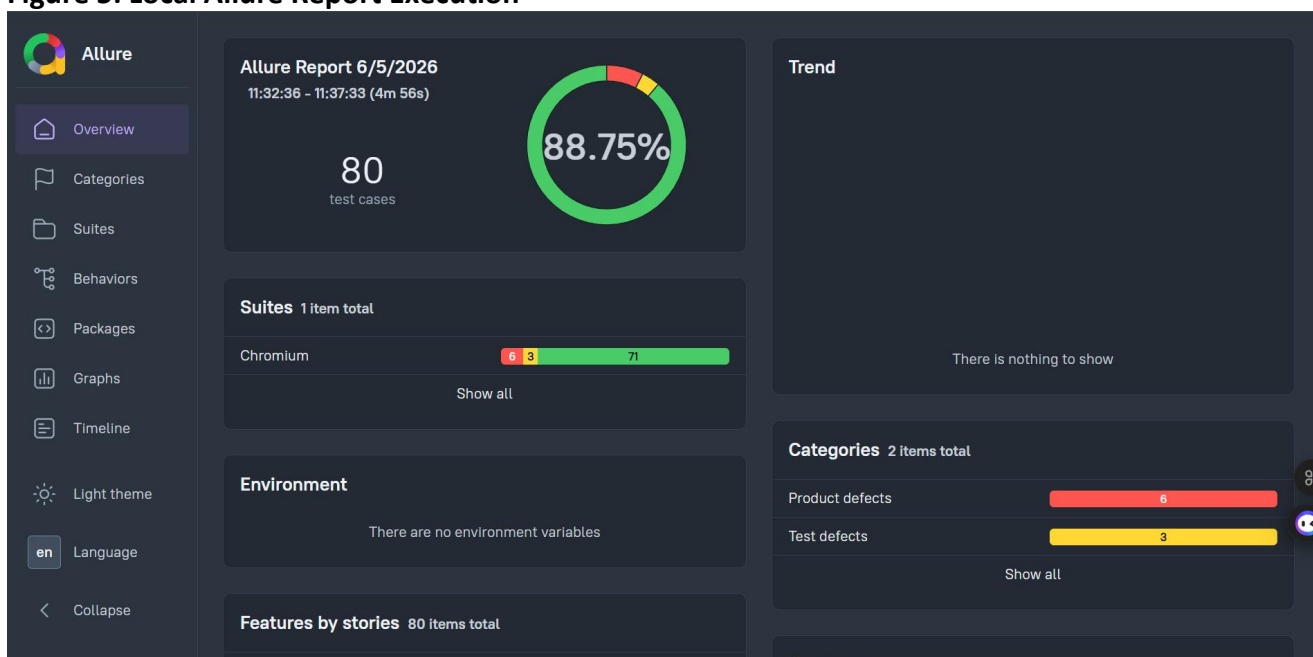
Figure 4: Allure Graphs and Analytics



This section displays graphical insights such as test execution trends, severity distribution, and timing analysis. It helps in understanding overall test stability and performance.

Local Allure Report Execution

Figure 5: Local Allure Report Execution



This section shows the locally generated Allure report executed on the development environment. It provides detailed step-by-step test execution logs and results.

Execution Results Summary

The test suite was executed on the **Chromium browser only**. All test cases were successfully automated using Playwright and executed via GitHub Actions CI/CD pipeline.

Test reports were generated using **Allure Reports** and deployed through **GitHub Pages**, ensuring easy access and visualization of results.

CI/CD Pipeline (GitHub Actions)

1. Overview

This project implements Continuous Integration and Continuous Deployment (CI/CD) using **GitHub Actions**. The pipeline is designed to automatically execute automated test cases whenever code is pushed to the repository or a pull request is created.

The main objective of the CI/CD pipeline is to ensure continuous validation of the Playwright automation framework and maintain code quality throughout development.

2. CI/CD Workflow Execution Flow

The CI/CD pipeline follows a structured execution flow as given below:

- Code is pushed to the GitHub repository
 - GitHub Actions workflow is triggered automatically
 - Node.js environment is initialized in the runner
 - Project dependencies are installed
 - Playwright browsers are installed
 - Test cases are executed in headless mode
 - Test execution reports are generated using Allure
 - Reports are published for analysis
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3. Key Commands Used

The following commands are executed during the CI/CD pipeline:

- `npm install` – Installs project dependencies
 - `npx playwright install` – Installs required Playwright browsers
 - `npx playwright test` – Executes automated test cases
 - `npm run allure:generate` – Generates Allure test reports
 - `npm run allure:open` – Opens the Allure report locally for analysis
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4. Purpose of CI/CD Pipeline

The implementation of CI/CD provides the following benefits:

- **Automated Regression Testing:** Ensures all test cases are executed on every code change
 - **Faster Feedback Cycle:** Developers receive immediate results after commits
 - **Consistency:** Ensures tests run in a standardized environment
 - **Code Quality Assurance:** Detects issues early in the development lifecycle
 - **Continuous Validation:** Maintains stability of the automation framework
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5. Conclusion

The CI/CD pipeline built using GitHub Actions enhances the reliability and efficiency of the Playwright automation framework. It ensures that every code update is automatically tested, validated, and reported, reducing manual effort and improving overall software quality.

Key Achievements

- Automated UI testing using Playwright (Chromium browser)
 - Implementation of Page Object Model (POM) architecture
 - Modular test structure for IMDb services
 - Integration of Allure reporting for test visualization
 - CI/CD pipeline setup using GitHub Actions
 - Automated report generation and deployment via GitHub Pages
 - Structured test coverage across multiple IMDb modules
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Conclusion

This capstone project demonstrates a scalable test automation framework built using **Playwright with JavaScript**. The framework follows the Page Object Model design pattern and integrates CI/CD pipelines for continuous testing.

Currently, execution is limited to the **Chromium browser**, but the architecture supports future expansion to cross-browser testing and enhanced test coverage. The project successfully demonstrates modern QA practices including automation, reporting, and continuous integration.